

신경재생 및 퇴행(이) (BR403-01)

학기	2026년 1학기
담당교수	조용철
이수구분	심화필수
학점	3.0
강의시간	화14:30-16:00(E7 - L21), 목14:30-16:00(E7 - L21)
강의실	
소속	기초학부

교과개요

● This course explores the biological mechanisms that govern the response of neural tissues to physical or physiological injury. Students will examine how neurons react through specific signaling cascades and gene expression changes, and how these processes determine whether neural circuits regenerate or degenerate. The course investigates the molecular and cellular principles underlying the differential regenerative capacities of the central and peripheral nervous systems, while tracing the historical development of neuroregeneration research over the past century. ● Building on this foundation, the course introduces major advances in neural repair and regeneration, including the roles of stem cells, growth factors, gene therapy, and tissue engineering approaches. The translational relevance of these studies to clinical and therapeutic applications will be emphasized. ● In addition, the course addresses the pathophysiological mechanisms of major neurodegenerative diseases such as Alzheimer's disease, Parkinson's disease, and amyotrophic lateral sclerosis (ALS). Students will explore current strategies aimed at preventing or reversing neuronal degeneration and promoting functional recovery in damaged neural systems.

교과목표

● Explain the cellular and molecular mechanisms that mediate neural responses to injury. ● Compare the regenerative capacities of the central and peripheral nervous systems and understand the biological principles underlying their differences. ● Interpret key findings and milestones from the past century of research on neural regeneration and degeneration. ● Evaluate recent experimental and therapeutic approaches, including stem cell therapy, gene therapy, and neurotrophic factor-based treatments. ● Analyze the pathogenesis of major neurodegenerative diseases and discuss emerging strategies to counteract neuronal loss. ● Integrate fundamental concepts to envision potential applications of neural repair and neuroprotection in clinical and biomedical contexts. ● Understand the fundamentals of multi-omics data, including genomics, transcriptomics, and proteomics, and apply basic data analysis techniques relevant to neuroregeneration and neurodegeneration research. ● Utilize AI-based tools for data visualization and interpretation to explore complex biological datasets and to generate integrative insights into neural function, damage, and recovery mechanisms.

주차별 강의계획

1주차:● 1주차 Topic: Introduction to Neural Regeneration and Degeneration. Contents: Course overview, objectives, and evaluation methods. Historical background of neurobiology and key discoveries shaping the field.
2주차:● 2주차 Topic: Neural Injury and Cellular Responses. Contents: Types of neural injury; mechanisms of neuronal death, axonal damage, and glial activation.
3주차:● 3주차 Topic: Signal Transduction and Gene Regulation in Neural Injury. Contents: Major intracellular signaling pathways and transcriptional control following neural damage.
4주차:● 4주차 Topic: Axon Growth and Guidance Mechanisms. Contents: Molecular cues, growth cone dynamics,

and extracellular matrix interactions in axonal regeneration.

5주차:● 5주차 Topic: Regeneration in the Peripheral Nervous System (PNS). Contents: Mechanisms and success factors in PNS repair; Schwann cell roles and trophic support.

6주차:● 6주차 Topic: Regeneration in the Central Nervous System (CNS). Contents: Barriers to CNS regeneration: glial scar formation, inhibitory molecules, and neuronal plasticity.

7주차:● 7주차 Topic: Stem Cells and Neural Repair. Contents: Neural stem cell biology, differentiation, transplantation, and regenerative applications.

8주차:● 8주차 Topic: No Class (Midterm Break)

9주차:● 9주차 Topic: Gene Therapy and Growth Factors. Contents: Gene delivery systems, neurotrophic factors (NGF, BDNF, GDNF), and modulation of regenerative pathways.

10주차:● 10주차 Topic: Multi-omics in Neuroscience. Contents: Overview of genomics, transcriptomics, proteomics, and metabolomics; data integration and analysis principles.

11주차:● 11주차 Topic: AI-Based Data Analysis and Visualization. Contents: Application of computational and AI tools for multi-omics data visualization and interpretation in neuroscience research.

12주차:● 12주차 Topic: History and Concepts of Neurodegeneration. Contents: Evolution of neurodegeneration research; major discoveries, hypotheses, and disease classification.

13주차:● 13주차 Topic: Molecular and Cellular Mechanisms of Neurodegeneration. Contents: Mechanisms of neuronal loss: oxidative stress, mitochondrial dysfunction, protein aggregation, and neuroinflammation.

14주차:● 14주차 Topic: Major Neurodegenerative Diseases (Alzheimer's, Parkinson's, ALS). Contents: Pathological features, molecular mechanisms, biomarkers, and current therapeutic approaches.

15주차:● 15주차 Topic: Current Research and Translational Approaches. Contents: Discussion of recent studies: regenerative medicine, neuroprotection, tissue engineering, and clinical implications.

16주차:● 16주차 Final Examination

교과정책(방침)

Students must have completed 'Cell Biology(세포생물학)', 'General Biology I(일반생물학1)', 'General Biology II(일반생물학2)', 'Advanced Brain Science I' OR 'Advanced Brain Science II' before enrolling in this course.